

EVALUATION OF RL-BASED ADAPTIVE TRAFFIC SIGNAL CONTROL UNDER DIFFERENT CONNECTED AND AUTONOMOUS VEHICLE PENETRATION LEVELS IN MIXED TRAFFIC

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ABSTRACT

Connected and Autonomous Vehicles (CAVs) and Reinforcement Learning (RL)-based Adaptive Traffic Signal Control (ATSC) systems have emerged as promising technologies for improving urban traffic operations. This study evaluates the performance of an RL-based ATSC system under varying traffic conditions, demand patterns, and CAV penetration levels in mixed traffic. Microscopic traffic simulation was conducted using SUMO in a synthetic Grid 4×4 network with 16 signalized intersections. Three traffic conditions were considered: the HDV condition (100% HDVs), mixed condition (50% HDVs and 50% CAVs), and CAV condition (100% CAVs). An Independent Deep Q-Network (IDQN) model was trained and evaluated across six traffic demand cases representing morning peak, roadblocks, evening peak, special events, adverse weather, and a full-day combined scenario, then compared against a conventional FixedTime controller using average travel time and recovery time. The IDQN model consistently outperformed the FixedTime controller across all conditions, achieving travel time reductions of up to 30.6% under the HDV condition and 34.3% under the CAV condition. Recovery times ranged from 1 to 8 hours, with the longest adaptation observed under roadblock and special event scenarios. The CAV condition consistently yielded the lowest optimum travel times, demonstrating that integrating CAV technology with RL-based ATSC significantly amplifies network performance. These findings confirm that RL-based ATSC systems effectively adapt to diverse and unseen traffic conditions while consistently outperforming fixed-time approaches, with benefits further amplified at higher CAV penetration levels.

Keywords: Reinforcement Learning, Adaptive Traffic Signal Control, Connected and Autonomous Vehicles, Mixed Traffic, Deep Q-Network, Recovery Time, SUMO

1. INTRODUCTION

Urban traffic congestion continues to impose severe economic, environmental, and safety costs on cities worldwide, and inefficient signal control is a major contributing factor [1]. Adaptive Traffic Signal Control (ATSC) systems that respond to real-time traffic states offer a practical way to reduce congestion without costly infrastructure expansion. Among ATSC approaches, Reinforcement Learning (RL) has emerged as a particularly promising method, as RL-based controllers learn signal-timing policies directly from interaction with the traffic environment and have been shown to outperform conventional fixed-time and actuated control strategies [2,3].

In parallel, Connected and Autonomous Vehicles (CAVs) are expected to reshape urban traffic operations through vehicle-to-vehicle and vehicle-to-infrastructure communication, enabling smoother car-following behaviour and reduced stop-and-go disturbances compared with Human-Driven Vehicles (HDVs) [4,5]. Because full CAV adoption will take many years, HDVs and CAVs are expected to coexist on the same road network for an extended transition period [6]. This raises an important question for traffic engineers: can RL-based ATSC maintain reliable, high-quality performance across this transition, under diverse and sometimes unseen traffic conditions and varying CAV penetration levels?

Despite extensive research on RL-based ATSC, relatively few studies have systematically examined how such controllers adapt when traffic demand patterns and vehicle-automation levels change simultaneously, and how quickly a trained RL agent recovers optimal performance after being exposed to a previously unseen scenario. This gap is significant because real-world RL-based controllers will

inevitably encounter traffic conditions that differ from their training data, and their ability to adapt directly affects the practicality of field deployment [7].

This paper addresses this gap by analysing the performance of an RL-based ATSC system under varying traffic conditions and patterns at different penetration levels of CAVs. An Independent Deep Q-Network (IDQN) controller is trained and evaluated in a synthetic Grid 4×4 network under three vehicle-composition conditions (HDV, mixed, and CAV) and six traffic demand cases, and its performance is benchmarked against a conventional FixedTime controller using average travel time and a newly introduced recovery-time metric.

2. METHODOLOGY

2.1 Simulation Environment and Vehicle Modelling

Microscopic traffic simulation was performed in SUMO (Simulation of Urban Mobility) using a synthetic Grid 4×4 network consisting of 16 signalized intersections (Figure 1). Horizontal roads are four-lane major roads and vertical roads are two-lane minor roads, with a standard link length of 200 m (75 m at the network boundary).

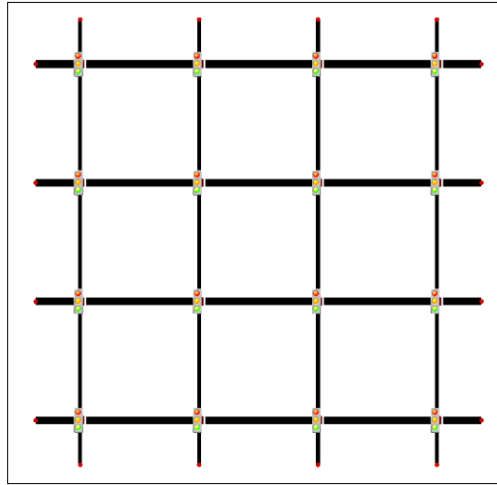


Figure 1: Grid 4×4 network with 16 signalized intersections used for simulation

HDVs were modelled using the Intelligent Driver Model together with driver reaction-time and perception-error parameters, while CAVs were modelled using a field-calibrated Cooperative Adaptive Cruise Control (CACC) car-following model with four control modes (speed control, gap control, gap-closing control, and collision avoidance) [8,9]. The default SUMO lane-changing model (LC2013) was used for both vehicle types. The calibrated car-following parameters used for the two vehicle types are summarized in Table 1.

Table 1: Car-following model parameters for the simulated HDVs and CAVs

Parameter	HDV	CAV
Maximum acceleration	1.0 m s ⁻²	2.9 m s ⁻²
Deceleration	2.0 m s ⁻²	7.5 m s ⁻²
Emergency deceleration	–	9.0 m s ⁻²
Time headway	1.5 s	0.6 s
Minimum gap	2.0 m	1.5 m
Free-flow speed	33.3 m s ⁻¹	–
Speed imperfection / deviation	0.5 (σ)	0.05

Three vehicle-composition conditions were considered to represent the transition from a fully human-driven fleet to a fully autonomous fleet: the HDV condition (100% HDVs), the mix condition (50% HDVs and 50% CAVs), and the CAV condition (100% CAVs), as summarized in Table 2.

Table 2: HDV and CAV penetration levels for the three traffic conditions

Condition Name	HDV	CAV
HDV Condition	100 %	0 %
Mixed Condition	50 %	50 %
CAV Condition	0 %	100 %

For each vehicle-composition condition, six traffic demand cases were developed using SUMO's `randomTrips.py` tool to represent a range of realistic and challenging traffic conditions. Case 1 represents morning peak-hour traffic, where trip origins are distributed across the network and most destinations are concentrated along the central roads representing the CBD, reflecting commuter movement toward the city centre. Case 2 builds on Case 1 by introducing two blocked roads to represent a sudden reduction in road capacity during peak demand. Case 3 represents evening peak-hour traffic, with the origin-destination pattern reversed from Case 1 to reflect the evening commute from the CBD toward the residential areas at the network edges. Case 4 extends Case 3 by adding special-event traffic, with vehicles travelling from different parts of the network toward the southern edge. Case 5 considers the same evening traffic conditions as Case 3 under adverse wet-road conditions, represented by a friction factor of 0.65 [10]. Finally, Case 6 represents a 14-hour operating period that combines morning traffic (hours 0-4, as in Case 1), special-event traffic (hours 6-9, as in Case 4), and evening traffic (hours 10-14, as in Case 3), while random traffic is maintained throughout the simulation. This case is intended to evaluate the controller's adaptability to a full operating period with changing and previously unseen traffic conditions. Cases 1-5 were each simulated for one hour. Overall, this design resulted in 18 distinct evaluation cases, consisting of three vehicle-composition conditions and six traffic demand cases, for the Grid 4×4 network.

2.2 RL-Based ATSC Model and Baseline Controller

Reinforcement learning (RL) is a machine-learning technique in which an agent learns a decision-making policy through repeated interaction with an environment (Figure 2): at each time step the agent observes a state S_t , selects an action A_t , and receives a reward R_{t+1} together with the next state S_{t+1} , with the goal of maximizing long-term cumulative reward. IDQN applies Q-learning, using a deep neural network to approximate the state-action value function, which allows it to handle the large, dynamic state and action spaces associated with traffic signal control; at each intersection, the agent evaluates the expected reward of each candidate signal phase and selects the phase expected to yield the best traffic performance.

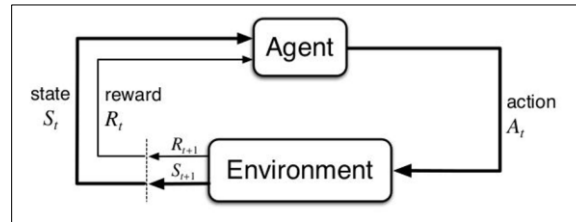


Figure 2: Agent-environment interaction loop in reinforcement learning

Two traffic signal control models were evaluated using this framework. In IDQN, a separate DQN agent is deployed at each of the 16 intersections; each agent observes only its local intersection state and is updated independently, without exchanging information with neighbouring intersections. FixedTime, used as the baseline, is a conventional non-RL controller that allocates a fixed green-time duration to each phase and represents standard real world signal control practice. Both controllers were implemented using the LibSignal RL traffic control testing framework, and all experiments were run in SUMO. The LibSignal repository link is provided here: <https://github.com/DaRL-LibSignal/LibSignal>.

2.4 Performance Metrics and Data Collection

Performance was quantified using two metrics: (i) average travel time, recorded per simulated episode for each case, and (ii) recovery time, defined in this study as the time required for the IDQN model to adapt to a previously unseen traffic pattern or condition and stabilize at its optimal performance level.

Because the episode wise travel time curve fluctuates considerably during learning, recovery time was estimated using a window-based analysis with predefined convergence thresholds, following the LibSignal evaluation procedure.

The IDQN model was first trained on Case 1 of the HDV condition for 200 episodes (each episode representing one hour of simulation); the trained weights were then used, without further case-specific retraining, to evaluate performance across all six cases of the HDV, mix, and CAV conditions. The FixedTime controller required no training and was evaluated directly under the same eighteen cases.

The Grid 4×4 scenario files can be accessed here: https://github.com/mdaslamcee/row_data_SUMO/tree/main/grid4x4. Data were collected from the simulation outputs of SUMO. Because the raw simulation outputs could not be used directly for evaluation, they were further processed and organized using customized scripts, available at https://github.com/mdaslamcee/RL_robust_data.

3. RESULTS AND DISCUSSION

3.1 Initial Training and Recovery Time

During initial training on HDV Case 1, the IDQN model required approximately 23 hours (simulated episodes) to stabilize, converging to an average travel time of 135.6 s. When the trained model was subsequently exposed to the eighteen evaluation cases, recovery times ranged from 1 to 8 hours, as summarized in Table 3.

Table 3: Case-wise recovery time (hours) of the IDQN model in the Grid 4×4 network

Cases	HDV Condition	Mix Condition	CAV Condition
Case 1 – Morning peak	5	6	6
Case 2 – Morning peak + roadblock	7	8	8
Case 3 – Evening peak	5	6	5
Case 4 – Evening peak + special event	8	6	6
Case 5 – Evening peak + adverse weather	5	6	6
	1	1	1

Cases 1, 3, and 5 exhibited moderate and consistent recovery times of 5-6 hours, indicating smooth adaptation to typical peak-hour and weather-affected demand. Case 2 (roadblock) and Case 4 (special event) required the longest recovery, up to 7-8 hours, suggesting that sudden capacity reductions and demand surges pose a greater adaptation challenge for the RL agent. In contrast, Case 6 stands out from the other cases due to its exceptionally fast convergence. Although the reduction from initial to optimum travel time is very small, this does not indicate poor performance. Rather, it shows that the pre-trained IDQN model was already well suited to the traffic conditions in this case, requiring very little additional learning. The consistently low travel times and shortest recovery times across all three conditions demonstrate the strong generalization capability of the trained model.

3.2 Travel Time Reduction and Comparison with FixedTime

Across all eighteen cases, the IDQN model consistently converged to a lower average travel time than the FixedTime controller. For example, in Case 1 under the HDV condition, IDQN converged to 130.8 s versus 208.25 s for FixedTime, and under the CAV condition to 68.3 s versus 101.24 s for FixedTime. The Table 4 presents all case-wise data.

The percentage reduction in travel time achieved by IDQN, measured from its initial to its optimum value, is summarized in Figure 3. The largest reductions occurred in Case 4 (evening peak with special-event traffic), reaching 30.6% under the HDV condition, 33.7% under the mix condition, and 34.3% under the CAV condition, the highest reduction observed among all cases. Case 2 (roadblock) produced the smallest reductions (16.2-22.0%), consistent with its longer recovery times. Case 6 showed only marginal reductions (0.3-1.9%) because the model was already operating close to its optimum from the first episode, again reflecting strong transfer of the learned policy rather than poor performance.

A consistent trend was also observed with respect to CAV penetration: for Cases 1, 3, and 5, the CAV condition achieved the lowest optimum travel times among the three conditions, followed by the mix condition and then the HDV condition. This indicates that the CAV condition achieved the lowest optimum travel times, followed by the mix condition and the HDV condition, indicating the benefits of higher CAV penetration when combined with adaptive signal control. Overall, the findings confirm that IDQN-based ATSC, particularly under higher CAV penetration, can significantly improve network travel time performance.

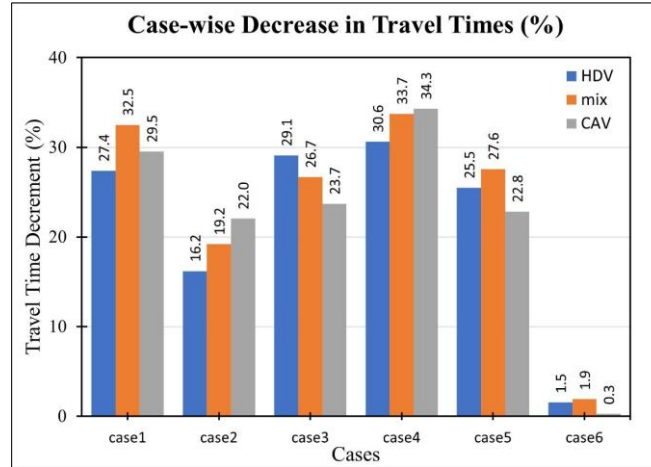


Figure 3: Case-wise percentage decrease in average travel time achieved by the IDQN model relative to its own initial performance, for the HDV, mix, and CAV conditions

Table 4: Case-Wise Initial and Optimum Average Travel Times (s), Percentage Reduction in Travel Times (%) for the IDQN and Fixed Travel Times (s) for the FixedTime Models under HDV, Mix, and CAV Conditions in the Grid 4×4 Network.

Cases	Traffic Condition	IDQN			FixedTime
		Initial Travel Time (sec)	Optimum Travel Time (sec)	Percent Decrease in Travel Times (%)	Fixed Travel Time (sec)
Case1	HDV	180.1	130.8	27.4	208.25
	mix	147.4	99.5	32.5	146.53
	CAV	96.9	68.3	29.5	101.24
Case2	HDV	209.4	175.5	16.2	239.81
	mix	172.5	139.4	19.2	183.31
	CAV	141.6	110.4	22.0	140.42
Case3	HDV	183.5	130.1	29.1	194.62
	mix	134.2	98.4	26.7	137.78
	CAV	89.9	68.6	23.7	97.66
Case4	HDV	207.6	144	30.6	208.43
	mix	166.6	110.4	33.7	164.48
	CAV	111.1	73	34.3	112.47
Case5	HDV	176.6	131.6	25.5	196.19
	mix	143	103.6	27.6	154.77
	CAV	97.7	75.4	22.8	102.18
Case6	HDV	136	133.9	1.5	250.91
	mix	105.1	103.1	1.9	149.80
	CAV	68.3	68.1	0.3	99.06

4. CONCLUSION

This study investigated the performance of an IDQN-based RL-ATSC system under different traffic conditions and demand patterns at three CAV penetration levels using a synthetic Grid 4×4 network. The IDQN model consistently achieved lower average travel times than the conventional FixedTime controller across all eighteen evaluated cases, with maximum reductions of 30.6% under the HDV condition and 34.3% under the CAV condition. Recovery times varied from 1 to 8 hours, with scenarios involving roadblocks and special events requiring longer adaptation, while the combined multi-pattern case stabilized rapidly, demonstrating the strong generalization capability of the learned policy. The CAV condition consistently resulted in the lowest optimum travel times, indicating that the integration of CAV technology with RL-based adaptive signal control can provide greater network-level benefits than either technology alone. These findings demonstrate the potential of RL-based ATSC for improving traffic operations in mixed CAV-HDV environments while highlighting the importance of developing training strategies that enhance robustness to sudden and previously unseen traffic disruptions. Future research should evaluate the proposed approach on larger real-world networks and conduct comparative assessments of multiple RL algorithms within a common evaluation framework.

ACKNOWLEDGEMENTS

The authors gratefully acknowledge the guidance and supervision of the Department of Civil Engineering, Bangladesh University of Engineering and Technology (BUET), in the development of this study.

GENERATIVE AI USE DISCLOSURE

Claude (Anthropic) and ChatGPT were used for paraphrasing. All AI-assisted text was reviewed, verified, and approved by the authors prior to submission.

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